DURATION : 180 Minutes | DATE : 16/03/2025 | M.MARKS : 198

ANSWER KEY

PHYSICS

1. (B)
2. (D)
3. (B)
4. (C)
5. (B)
6. (B)
7. (A, D)
8. (A, B, C)
9. (A, B, D)
10. (C, D)
11. (A, C)
12. (B, C)
13. (5.00)
14. (10.50)
15. (129.60)
16. (9.00)
17. (5.00)
18. (2.50)

CHEMISTRY

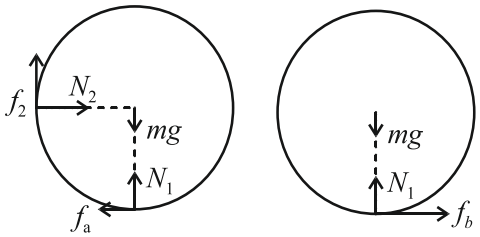
19. (C)
20. (A)
21. (C)
22. (D)
23. (A)
24. (B)
25. (A, B, D)
26. (A, C, D)
27. (C, D)
28. (A, B)
29. (A, B, C, D)
30. (A, B, C, D)
31. (28.00)
32. (0.69)
33. (5.76)
34. (49.20)
35. (96.47)
36. (12.50)

MATHEMATICS

37. (B)
38. (B)
39. (B)
40. (C)
41. (B)
42. (B)
43. (B, C)
44. (A, B, C, D)
45. (A, B, C)
46. (A, B, C)
47. (A, B, C, D)
48. (A, D)
49. (0.00)
50. (2.00)
51. (8.00)
52. (0.50)
53. (30.00)
54. (12.00)

PART-I (PHYSICS)

1. (B)



$f_2 + N_1 = mg$

$f_a = N_2 = \mu N_1$

$f_a = \mu\,(mg - f_2)$

$f_a = \mu\,[mg - \mu N_2]$

$f_a = \mu\,[mg - \mu f_a]$

$f_a\,(1 + \mu^2) = \mu\, mg$

$$f_a = \frac{\mu mg}{1+\mu^2} = \frac{3mg}{10}$$

$f_b = \mu N_1 = \mu mg$

$$f_b = \frac{mg}{3}$$

$$\frac{f_a}{f_b} = \frac{9}{10}$$

2. (D)

During each elastic collision the balls will exchange velocities. It is given that all the balls are identical (the two colliding balls have no names), We could say that the balls simply pass through each other moving independent of each other. If a ball was at the edge it will take longest time to fall off the other edge.

$$\therefore\ t_{max} = \frac{L}{v}$$

3. (B)

Due to radial symmetry

$$E_A = \frac{1}{4\pi\varepsilon_0}\frac{\int_0^x 4\pi x^2 dx \times \rho_0 x^2}{x^2}$$

$$\Rightarrow E_A = \frac{1}{4\pi\varepsilon_0} 4\pi\rho_0 \frac{x^3}{5}$$

and $$E_B = \frac{1}{4\pi\varepsilon_0}\frac{\int_0^R 4\pi x^2 dx \times \rho_0 x^2}{y^2}$$

$$\Rightarrow E_B = \frac{1}{4\pi\varepsilon_0} 4\pi\rho_0 \frac{R^5}{5y^2}$$

Equating E_A and E_B we get

$$\frac{1}{4\pi\varepsilon_0} 4\pi\rho_0 \frac{x^3}{5} = \frac{1}{4\pi\varepsilon_0} 4\pi\rho_0 \frac{R^5}{5y^2}$$

$\Rightarrow x^3y^2 = R^5$

4. (C)

When r_v is large,

$$16 = \left(\frac{20}{R+r}\right)R \qquad \ldots.(1)$$

When $r_v = 8$,

$$R' = \frac{8R}{8+R}$$

$$\frac{40}{3} = \frac{20R'}{r+R'} \qquad \ldots.(2)$$

Solving (1) & (2)

$r = 2\Omega$, $R = 8\Omega$

5. (B)

Let in new equilibrium upper plate is displaced by x units.

If we get real x less than L then only new equilibrium is achieved.

Here, force between plates of capacitor = spring force

$Kx = \dfrac{Q^2}{2A\varepsilon_0}$, where x = extension in spring and A is area of the plate.

We know that, after connecting the switch,

$Q = C'V$

So, $$Kx = \frac{1}{2A\varepsilon_0}\left[\frac{A\varepsilon_0}{L-x}V\right]^2$$

Maximum value of $Kx\,(L-x)^2$ for V_{max}

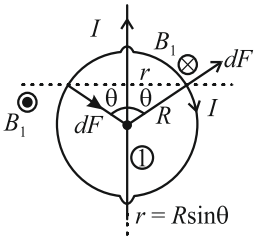
We get $K(L-x)^2 + 2Kx(L-x)(-1) = 0 \quad x = L/3$

Maximum value of $\quad Kx(L-x)^2 = K\dfrac{4L^3}{27}$

So for real x,

$$K\frac{4L^3}{27} \geq \frac{A\varepsilon_0}{2}V^2 \quad \Rightarrow V \leq \sqrt{\left(\frac{2}{3}\right)^3 \frac{L^3K}{A\varepsilon_0}}$$

6. (B)



$dF_{res} = 2\,(dF)\sin\theta$

$= 2I\,(d\ell)\,B\sin\theta$

$$= 2I(Rd\theta)\left(\frac{\mu_0 I}{2\pi r}\right)\sin\theta$$

$$= \frac{\mu_0 I^2 R}{\pi}\frac{\sin\theta d\theta}{(r)}$$

$$= \frac{\mu_0 I^2 R}{\pi}\frac{\sin\theta}{(R\sin\theta)}d\theta = \frac{\mu_0 I^2}{\pi}(d\theta)$$

$$F_{res} = \int dF_{res} = \frac{\mu_0 I^2}{\pi}\int_0^{\pi} d\theta = \mu_0 I^2$$

7. (A, D)

Initial velocity of efflux is

$$v = \sqrt{2gh} = \sqrt{2\times 10\times\frac{5}{100}}$$

$\Rightarrow$ v = 1 m/s

Range, $R = u_x T = v\cos\theta\times\frac{1}{5}\left(T = \frac{1}{5}\sec\right)$

Range from $Q = (R - 6)$ cm = 0.10 m

Horizontal momentum of the water coming out per second of the hole

$F = A\rho v^2\cos\theta = 1.5\times 10^{-4}\times 10^3\times 1^2\times 0.8$

$\Rightarrow$ F = 0.12

8. (A, B, C)

$$V = L\frac{di}{dt} = 2t$$

$$\Rightarrow \quad i = \frac{t^2}{L}$$

$$U = \frac{1}{2}Li^2 = \frac{1}{2}\times 2\times\left(\frac{2^2}{2}\right)^2$$

$\Rightarrow$ U = 4 J

$$U = \frac{1}{2}Li^2 = \frac{1}{2}L\frac{t^4}{L^2}$$

$$\Rightarrow \quad U = \frac{t^4}{2L}$$

$$\Rightarrow \quad \left.\frac{dU}{dt}\right|_{t=1} = 2\frac{t^3}{L} = 1\text{ J/s}$$

9. (A, B, D)

Intensity of gravitational field due to long hollow cylinder can be found using Gauss law, field and hence Force will be inversely proportional to radius of the orbit and directly proportional to mass per unit length of the planet.

$$F = \frac{2GM'm}{r}$$

$$\frac{mv^2}{r} = \frac{2GM'm}{r} = \frac{2Gm\sigma 2\pi R}{r}$$

$\Rightarrow$ $KE \propto R$

$\Rightarrow$ $KE \propto r^0$

$\Rightarrow$ $v \propto r^0$

$$T = \frac{2\pi r}{v}$$

$\Rightarrow$ $T \propto r$

10. (C, D)

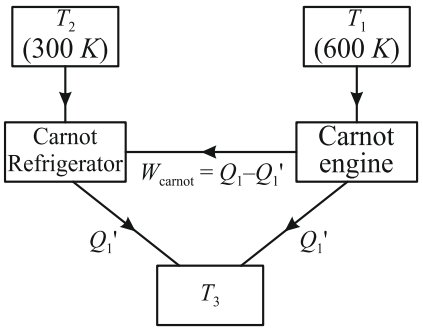
Least count of device = $\frac{1}{30}\left(\frac{1}{2}\right)^{\circ} = 1'$

δ_R = 30°16' – 20°20' = 9°56'

δ_γ = 32°39' – 20°20' = 12°19'

δ_v = 33°41' – 20°20' = 13°21'

11. (A, C)



$$\eta_{\text{carnot engine}} = \frac{Q_1 - Q_1'}{Q_1} = \frac{T_1 - T_3}{T_1}$$

$$= \frac{\text{Work of Carnot engine}}{\text{Heat supplied to the Carnot engine}} = \frac{W_{\text{carnot}}}{Q}$$

$$W_{carnot} = Q_2\left(\frac{T_3 - T_2}{T_2}\right)\ldots \text{(ii)}$$

(i) Temperature, T_3:

From equations (i) and (ii), we get

$$Q_1\left(\frac{T_1 - T_3}{T_1}\right) = Q_2\left(\frac{T_3 - T_2}{T_2}\right)$$

$$\therefore \frac{Q_2}{Q_1} = \frac{T_2}{T_1}\left(\frac{T_1 - T_3}{T_3 - T_2}\right) \text{ or } \frac{Q_2}{Q_1} = 1 = \frac{300}{600}\left(\frac{600 - T_3}{T_3 - 300}\right)$$

or 600 – T_3 = 2 (T_3 – 300) or 600 – T_3 = 2 T_3 – 600

or T_3 = 400 K

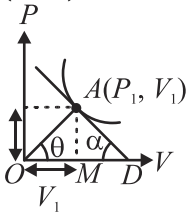
(ii) Efficiency of Carnot engine,

$$\eta_{\text{carnot engine}} = \frac{T_1 - T_3}{T_1}$$

$$= \frac{600 - 400}{600} = 0.3333 = 33.33\%$$

$$\text{COP}_{\text{refrigerator}} = \frac{T_2}{T_3 - T_2} = \frac{300}{400 - 300} = 3$$

12. (B, C)



$\tan\alpha = -$ [slope at (P_1, V_1)] $= \dfrac{P_1}{V_1}$

$\tan\theta = \dfrac{P_1}{V_1}$

$\therefore \quad OM = MD = V_1$

$\therefore \quad OD = 2V_1$

Area of triangle $AOD = \dfrac{1}{2}(AM \times OD)$

$= \dfrac{1}{2}(P_1 \times 2V_1) = P_1V_1 = nRT$

13. (5.00)

$T = 2t = 10$

$\Rightarrow \quad t = 5$ s

$t = \dfrac{10}{v_{my}} = 5$

$\Rightarrow \quad v_{my} = 2$ m/s

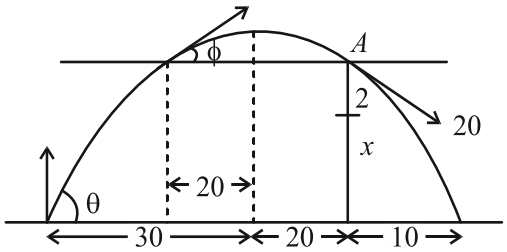
Horizontal drift at river middle

$x = (v_{mx} - 10)t = (20 - v_{mx})t$

$\Rightarrow \quad v_{mx} = 15$ m/s

$\tan\theta = \dfrac{v_{mx}}{v_{my}} = \dfrac{15}{2}$

14. (10.50)



Above the horizontal line $\leftrightarrow$, a different projectile is formed, $u = 20$, $R = 40$, by derivation

So, $40 = \dfrac{400 \times \sin 2\phi}{g}$

$\therefore \quad \phi = 45°$

$\therefore \quad 20\cos\phi = 10\sqrt{2} = u\cos\theta$

Time of projectile

$= \dfrac{R}{u_0 \cos\phi} = \dfrac{40}{10\sqrt{2}} = 2\sqrt{2}$

Now, for full trajectory, $R = 60$, $u\cos\theta = 10\sqrt{2}$

we can find time of full football trajectory

$T = \dfrac{R}{u\cos\theta} = 3\sqrt{2}$

$\therefore \quad$ To reach point A, it will take

$t = \dfrac{3\sqrt{2}}{2} + \dfrac{2\sqrt{2}}{2} = \dfrac{5}{\sqrt{2}}$

Now, in full football trajectory,

$T = 3\sqrt{2} = \dfrac{2u\sin\theta}{g} = \dfrac{u\sin\theta}{5}$

$\boxed{\therefore u\sin\theta = 15\sqrt{2}}$

By 2nd equation in vertical motion of football,

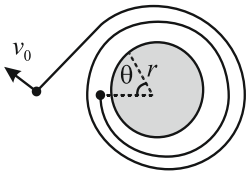
$S = 15\sqrt{2} \times \dfrac{5}{\sqrt{2}} - \dfrac{10}{2} \times \dfrac{25}{2}$

$= 75 - 62.5 = 12.5$

We can tell $s = 2 + x$

$\therefore \quad \boxed{x = 10.5 \text{ m}}$

15. (129.60)



$-dl = rd\theta$

$\dfrac{d\theta}{dt} = \dfrac{v_0}{l}$

$-\int_L^0 l\,dl = rv_0 \int_0^t dt$

$\dfrac{L^2}{2} = rv_0 t$

Time for unwinding, $t_1 = \dfrac{L^2}{2rv_0}$

then particle will move in a semicircle of radius

$L \Rightarrow t_2 = \dfrac{\pi L}{v_0}$

then time for wounding $t_3 = \dfrac{L^2}{2rv_0}$

$T = \dfrac{L^2}{rv_0} + \dfrac{\pi L}{v_0} = \dfrac{L}{rv_0}(L + \pi r)$

16. (9.00)

Energy of protonium atom $= \left(\dfrac{40}{3}\text{ eV}\right)\left(\dfrac{\mu}{m_e}\right)\dfrac{1}{n^2}$

Energy released in transition form $n = 2 \rightarrow n = 1$

$$= \frac{40}{3}\left(\frac{900\,m_e}{m_e}\right)\left(1-\frac{1}{4}\right) = 9000 \text{ eV}$$

17. (5.00)

$$v = v_0\left(\frac{c}{c-v}\right)$$

$$\Rightarrow \quad v = c\left(\frac{v-v_0}{v}\right)$$

$$t = \frac{\Delta x}{v} = \frac{\Delta x v}{c(v-v_0)}$$

$$\Rightarrow \quad t = \frac{78\times 275}{330\times(275-262)} = 5\text{s}$$

18. (2.50)

Torque due to magnetic force

= Torque due to weight

$$\Rightarrow \quad \int_0^{\pi} Bi(rd\theta)\sin\frac{\theta}{2}\times\left(2r\cos\frac{\theta}{2}\right)$$

$= mgr$

$$\Rightarrow \quad Bir^2\int_0^{\pi}\sin\theta d\theta = mgr$$

$$\Rightarrow \quad i = \frac{mg}{2Br} = \frac{100\times10\times100}{1000\times2\times2\times10} = 2.5 \text{ A}$$

PART-II (CHEMISTRY)

19. (C)

$T\ell I_3 \rightarrow T\ell^{+1} + I_3^-$

20. (A)

Boiling takes place when vapour pressure is equal to external pressure. If we are still heating then vapour pressure will remain same but temperature of the water increases.

21. (C)

(A) Theory based true statement.

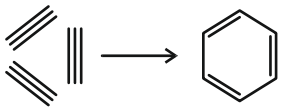
(B) Borax in aq. Solution gives Boric acid & its salt.

(C) B_2H_6 with excess NH_3 at low temperature gives $B_3N_3H_6$ which is aromatic but more reactive than benzene.

(D) Theory based true statement.

22. (D)

(A) $KE = \frac{15}{30}\times\frac{3}{2}\times RT = \frac{3}{4}RT$

(B) 

$$V = \frac{nRT}{P} = \frac{3RT}{P} = \frac{3RT}{P}$$

(D) $\frac{r_{H_2}}{r_{NH_3}} = \sqrt{\frac{M_{w_{NH_3}}}{M_{w_{H_2}}}} = \sqrt{\frac{17}{2}} = 2.92$

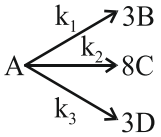
23. (A)

It occurs during roasting in reverberatory furnace at moderate temperature in presence of air.

24. (B)

No. of half life = $\frac{T}{t_{\frac{1}{2}}} = \frac{45}{15} = 3$

Mol of A remained after 45 days = $\frac{A_0}{2^n} = \frac{1}{8}$



$K_1 : K_2 : K_3 : 4 : 2 : 1$

Moles of A converted into product = $\frac{7}{8}$

Moles of [C] = $\frac{k_2}{k_1+k_2+k_3}\times\frac{7}{8}\times\frac{8}{1} = 2$

25. (A, B, D)

Conceptual

26. (A, C, D)

Al^{+3}, Sn^{+2}, Zn^{+2} and Pb^{+2} form soluble complex with NaOH.

27. (C, D)

$-NH_2 > -OH$ (+M effect)

$-NO_2 > -SO_3H$ (–M effect)

28. (A, B)

Solute X will dissociate into the solution so i > 1.

29. (A, B, C, D)

All the reasons given in the statements are correct and explains why nitrogen atom has low electron affinity.

30. (A, B, C, D)

Theory based.

31. (28.00)

	Combining orbitals (Internuclear axis)	Type of bond formed
(i)	$s + p_x(x)$	σ
(ii)	$d_{xy} + d_{xy}(x)$	π
(iii)	$d_{yz} + p_z(z)$	non-bonding
(iv)	$s + s(z)$	σ
(v)	$d_{yz} + d_{yz}(x)$	δ
(vi)	$p_y + p_y(y)$	σ
(vii)	$d_{z^2} + d_{z^2}(z)$	σ
(viii)	$d_{xy} + d_{xy}(z)$	δ
(ix)	$p_x + p_x(y)$	π
(x)	$s + p_z(x)$	non-bonding

32. (0.69)

$m_1 = 0.1$ kg, $m_2 = 0.012$ kg

T_b of sol. = 100.57°C T_b of water = 100°C.

Molar mas of $Ba(NO_3)_2 = 261.3$ g mol^{-1}

$\Delta T_b = K_b m \times i$

$$0.57 = 0.52 \times \frac{(0.012 / 0.2613)}{0.1} \times i$$

$i = 2.3868 \cong 2.38$

$i = 1 + (n - 1)\alpha$

$$\frac{2.38 - 1}{(3 - 1)} = \alpha$$

$\alpha = 0.69$

33. (5.76)

$$\Delta S_A = 0.5R\ell n\ \frac{2}{1}$$

= 2.88 Joule

Same for B, $\Delta S_B = 2.88$ Joule

$\therefore \quad \Delta S_{Total} = 2.88 + 2.88 = 5.76$ Joule

34. (49.20)

$$[\alpha] = \frac{\theta}{\ell \times c} = \frac{24.6}{\frac{2}{10} \times \frac{25}{10}}$$

= 49.20

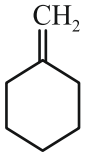
35. (96.47)

Percentage yield of 2-Bromopropane

$$= \frac{164}{170} \times 100$$

= 96.47

36. (12.50)

Product (R) is 

PART-III (MATHEMATICS)

37. (B)

Let P be (α, β, γ) so points A, B and C are $(\alpha, \beta, 0), (\alpha, 0, \gamma)$ and $(0, \beta, \gamma)$ respectively. So, plane passing through these points is $\frac{x}{\alpha} + \frac{y}{\beta} + \frac{z}{\gamma} = 2$ also $\frac{\alpha}{a} + \frac{\beta}{b} + \frac{\gamma}{c} = p$

If (x_1, y_1, z_1) is foot of perpendicular from origin, then equation of plane

$$x_1(x - x_1) + y_1(y - y_1) + z_1(z - z_1) = 0$$

Comparing with $\frac{x}{\alpha} + \frac{y}{\beta} + \frac{z}{\gamma} = 2$, we get

$$2\alpha x_1 = 2\gamma z_1 = x_1^2 + y_1^2 + z_1^2$$

So locus is

$$\left(x_1^2 + y_1^2 + z_1^2\right)\left(\frac{1}{ax_1} + \frac{1}{by_1} + \frac{1}{cz_1}\right) = 2p$$

38. (B)

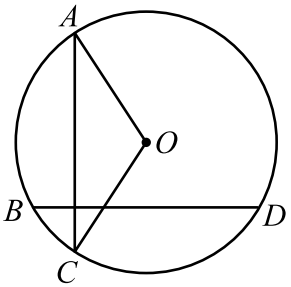
$$I = \int \tan^{-p/3} x \cdot \sec^4 x dx$$

$$= \int \tan^{-p/3} x(1 + \tan^2 x) \cdot \sec^2 x dx$$

39. (B)

If $a_1c_2 < 0$ and $a_2 c_1 < 0$ then both $D_1 > 0$ and $D_2 > 0$

40. (C)



As $OA = OC$

So, $\overrightarrow{OA}+\overrightarrow{OC}$ will be perpendicular to AC

$\therefore \quad \overrightarrow{OA}+\overrightarrow{OC}=\lambda\overrightarrow{BD}$

Where λ is the real number.

Similarly

$\overrightarrow{OB}+\overrightarrow{OD}=\mu\overrightarrow{AC}$

Where μ is the real number.

Now using (1) and (2), we get

$\left(\overrightarrow{OA}+\overrightarrow{OC}\right).\left(\overrightarrow{OB}+\overrightarrow{OD}\right)=\mu\cdot\lambda\,\overrightarrow{AC}\cdot\overrightarrow{BD}=0$

41. (B)

$P_r = 1 + \cos(2^{r-1}\theta) - \sin(2^{r-1}\theta) = 2\cos^2(2^{r-2}\theta) - 2\sin(2^{r-2}\theta)\cos(2^{r-2}\theta)$

$= 2\cos(2^{r-2}\theta) \times \{\cos(2^{r-2}\theta) - \sin(2^{r-2}\theta)\}$

$$Q_{r-1}=1+\tan\left(2^{r-2}\theta\right)=\frac{\cos\left(2^{r-2}\theta\right)+\sin\left(2^{r-2}\theta\right)}{\cos\left(2^{r-2}\theta\right)}$$

$$\therefore \quad P_rQ_{r-1}=2\cos\left(2^{r-1}\theta\right)$$

$\because \quad P_nQ_{n-1} = 2\cos 2^{n-1}\theta$

$\therefore \quad P_1Q_0 \times P_2Q_1 \times \ldots\ldots \times P_nQ_{n-1} = 1$

$$\frac{2^n\sin\theta\cos\theta\cos2\theta\cos2^2\theta....\cos2^{n-1}\theta}{\sin\theta}=1$$

$$\frac{\sin2^n\theta}{\sin\theta}=1$$

$\sin2^n\theta = \sin\theta$

42. (B)

$$\sum_{r=1}^{10}{}^{15}C_r\cdot{}^{35}C_{10-r}={}^{15}C_1\cdot{}^{35}C_9+{}^{15}C_2\cdot{}^{35}C_8\ldots.$$

$$....+{}^{15}C_{10}\cdot{}^{35}C_0$$

$$={}^{50}C_{10}-{}^{35}C_{10}$$

43. (B, C)

$\Delta_1 = 3abc - a^3 - b^3 - c^3$

Δ_2 is formed by cofactors of elements of Δ_1

$\Rightarrow \quad \Delta_2 = \Delta_1^2$

Also $\Delta_3 = \Delta_1$

44. (A, B, C, D)

For *A* and *D* option we use P.I.E

Option A: $\dfrac{3^6-{}^3C_1 2^6+3}{3^6}=\dfrac{20}{27}$

Option B : favourable cases

$$=\frac{6!}{5!}+\frac{6!}{2!\,3!}+\frac{6!}{4!}+\frac{6!}{3!\,3!}+\frac{6!}{3!\,2!}+\frac{6!}{5!}$$

Probability $=\dfrac{182}{3^6}$

Option C : favourable cases

$$=\frac{\frac{6!}{2!3!}+\frac{6!}{4!}+\frac{6!}{3!2!}}{3^6}=\frac{150}{3^6}$$

Option D: $\dfrac{{}^3C_2\left(2^6-2\right)}{3^6}=\dfrac{62}{243}$

45. (A, B, C)

Number of divisors of form $(4K + 2) = 1 \times 5 \times 3 \times 6 - 1 = 89$

Number of divisors of $n^2 = 7 \times 9 \times 5 \times 11$

Number of divisors of $n = 4 \times 5 \times 3 \times 6$

So, number of divisors of n^2, which are not divisors of n is $7 \times 9 \times 5 \times 11 - 4 \times 5 \times 3 \times 6 = 3105$

46. (A, B, C)

Let $\tan^{-1}x = \alpha$ and $\tan^{-1}x^3 = \beta$

$\tan\alpha = x$ and $\tan\beta = x^3$

$$\therefore \quad 2\tan(\alpha+\beta)=\frac{2(\tan\alpha+\tan\beta)}{1-\tan\alpha\tan\beta}$$

$$=2\left[\frac{x+x^3}{1-x^4}\right]=\frac{2x}{1-x^2}$$

Also,

$$\frac{2x}{1-x^2}=\frac{2\tan\alpha}{1-\tan^2\alpha}=\tan2\alpha=\tan\left(2\tan^{-1}x\right)$$

$$=\tan\left(2\left(\frac{\pi}{2}-\cot^{-1}x\right)\right)$$

$$=\tan\left(\pi-\cot^{-1}x-\cot^{-1}x\right)$$

$$=\tan\left(\cot^{-1}(-x)-\cot^{-1}(x)\right)$$

47. (A, B, C, D)

$$f'(x)=e^{-\frac{1}{x^2}}\cdot\frac{2}{x^3}$$

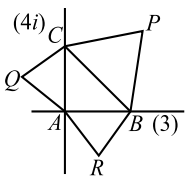
$$f''(x)=e^{-\frac{1}{x^2}}\times\left(\frac{4-6x^2}{x^6}\right)$$

$|f(x)| > f'(x)$

$$\Rightarrow \quad e^{-\frac{1}{x^2}}>e^{-\frac{1}{x^2}}\times\frac{2}{x^3}$$

$$\Rightarrow \quad x\in\left(\sqrt[3]{2},\infty\right)$$

48. (A, D)



Using rotation, we get complex number corresponding to *P*, *Q*, *R* as

$3+(3-4i)\omega$, $4i+4i\omega$ and -3ω respectively

So, centroid of $\Delta PQR=\dfrac{3+4i}{3}$

49. (0.00)

$f(x) \in (0, 1)$

$\lim_{n\to\infty}\left(f(x)\right)^n = 0$

50. (2.00)

Equation of ellipse can be written as $\frac{\left(\frac{x+y+1}{\sqrt{2}}\right)^2}{2}+\frac{\left(\frac{x-y-3}{\sqrt{2}}\right)^2}{1}=1$ which is similar to

$\frac{x^2}{2}+\frac{y^2}{1}=1$ whose director circle is $x^2 + y^2 = 3$

Let $P\equiv\left(\sqrt{3}\cos\theta,\sqrt{3}\sin\theta\right)$ equation of chord of contact is $x\sqrt{3}\cos\theta+2y\sqrt{3}\sin\theta-2=0$

It touches the circle $x^2 + y^2 = r^2$

$\Rightarrow r=\frac{|0+0-2|}{\sqrt{3\cos^2\theta+12\sin^2\theta}}=\frac{2}{\sqrt{12-9\cos^2\theta}}$

$\Rightarrow \quad r_{max}=\frac{2}{\sqrt{3}}$ and $r_{min}=\frac{2}{\sqrt{12}}\Rightarrow\frac{r_{max}}{r_{min}}=2$

51. (8.00)

I.F. = e^x

$\therefore$ solution is $f(x)\,e^x = \int 4x\sin 2x\,dx$

$\Rightarrow \quad f(x)\,e^x = 4\left(\frac{-x\cos 2x}{2}+\frac{\sin 2x}{4}\right)$

$\Rightarrow \quad f(x) = (-2x\cos 2x + \sin 2x)e^{-x}, f\left(\frac{\pi}{4}\right)=e^{-\frac{\pi}{4}}$

$\therefore \quad f\left(\frac{5\pi}{4}\right)=e^{-\frac{5\pi}{4}}$

$f\left(\frac{9\pi}{4}\right)=e^{-\frac{9\pi}{4}}$

$e^{-\frac{\pi}{4}}\left(\frac{e^{-8\pi}-1}{e^{-\pi}-1}\right)=\sum_{n=0}^{7} f\left(\frac{(4n+1)\pi}{4}\right)$

52. (0.50)

Trace $A=\sum_{i=1}^{n}(-1)^i\left(\frac{2i^2+1}{4i^4+1}\right)$

$\sum_{i=1}^{n}\frac{1}{2}(-1)^i\left(\frac{1}{2i^2-2i+1}+\frac{1}{2i^2+2i+1}\right)$

$\lim_{n\to\infty}\frac{1}{2}\sum_{i=1}^{n}\left(\frac{(-1)^i}{2i^2-2i+1}-\frac{(-1)^{i+1}}{2i^2+2i+1}\right)=-\frac{1}{2}$

53. (30.00)

Let $T_{k+1}=\frac{1}{a+kd}$ so $\frac{1}{a+5d},\frac{1}{a+9d},\frac{1}{a+17d}$ are in Arithmetic progression

$\Rightarrow \frac{1}{a+9d}-\frac{1}{a+5d}=\frac{1}{a+17d}-\frac{1}{a+9d}\Rightarrow a=7d$

$\Rightarrow T_p=\frac{1}{(p+6)d},T_{12}=\frac{1}{18d},T_q=\frac{1}{(q+6)d}$

$\Rightarrow 18^2=(p+6)(q+6)$ so maximum value of q is 30 (as $p+6\geq 7$)

So, minimum value of $p + 6 = 9$

$\Rightarrow q+6=36\Rightarrow q=30$

54. (12.00)

Let the coordinate of any point on Parabola be $(t^2 + 1, 2t)$

Let the reflection of $(t^2 + 1, 2t)$ in $x + y + 1 = 0$ be (h, k)

So, $\frac{h-\left(t^2+1\right)}{1}=\frac{k-2t}{1}=\frac{-2\left(t^2+1+2t+1\right)}{1+1}$

$\Rightarrow \quad h = -2t - 1$ and $k = -t^2 - 2$

So, on eliminating t

$h^2 + 2h + 4k + 9 = 0$

$\Rightarrow \quad x^2 + 2x + 4y + 9 = 0$

Kindly Share Your Feedback for This Paper



PW Web/App - https://smart.link/7wwosivoicgd4

Library- https://smart.link/sdfez8ejd80if